Faculty of Engineering & Technology

Electrical & Computer Engineering Department

Computer Networks Laboratory - ENCS4130

Report 1

Experiment No. 5

Path Vector Routing Protocols (Border Gateway Protocol - BGP)

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Abstract

This experiment focuses on configuring and verifying dynamic routing using the Border Gateway Protocol (BGP), a widely used exterior gateway protocol for interconnecting Autonomous Systems (AS) on the Internet. The work involves building a complete simulated environment in Cisco Packet Tracer using four routers, one switch, multiple PCs, and sniffer tools to analyze routing behavior. The experiment aims to develop a practical understanding of BGP's Path Vector mechanism, its integration with interior routing protocols such as OSPF and EIGRP, and its role in enabling scalable inter-domain communication. The main deliverables include establishing iBGP and eBGP peerings, performing route redistribution between BGP and IGPs, configuring timers, and analyzing routing updates and traffic flows using packet sniffing. This experiment provides hands-on experience with BGP configuration, neighbor verification, and route propagation within a controlled simulation environment.

Table of Contents

Abstract I
Acronyms and Abbreviations IV
Table of figures V
List of tables VI
1. Introduction 1
1.1. Border Gateway Protocol (BGP) 1
1.2. Using BGP 1
1.3. BGP Peers /Neighbors 2
1.4. BGP Peers Messages 2
1.5. BGP Finite-State Machine (FSM) 3
2. Procedure and Discussion 4
2.1. Building the Topology 4
2.2. Configuring OSPF Routing 6
2.3. Configuring BGP Routing 7
2.3.1. Configuring BGP Routing on Router 1 7
2.3.2. Configuring BGP Routing on Router 2 8
2.4. Define the BGP Over the OSPF 8
2.4.1. Define BGP Over OSPF in Router 1 9
2.4.2. Define BGP Over OSPF in Router 2 9
2.5. Define the OSPF Over the BGP 9
2.5.1. Define OSPF Over BGP in Router 1 10
2.5.2. Define OSPF Over BGP in Router 2 10
2.6. Configuring BGP Timers 11
3. Results 14
3.1. Viewing BGP Neighbors 14
3.2. Ping Between Devices 19
3.2.1. Ping Between PC1 (192.70.1.34) and PC3 (192.70.1.130) 19
3.2.2. Ping Between PC1 (192.70.1.34) and PC5 (192.70.1.194) 20
3.2.3. Ping Between PC2 (192.70.1.66) and PC3 (192.70.1.130) 20
3.2.4. Ping Between PC2 (192.70.1.66) and PC5 (192.70.1.194) 21

3.2.5. Ping Between PC3 (192.70.1.130) and PC1 (192.70.1.34) 21
3.2.6. Ping Between PC3 (192.70.1.130) and PC2 (192.70.1.66) 22
3.2.7. Ping Between PC5 (192.70.1.194) and PC1 (192.70.1.34) 22
3.2.8. Ping Between PC5 (192.70.1.194) and PC2 (192.70.1.66) 23
Conclusion 24
Feedback 25
References 26

Acronyms and Abbreviations

- **OSPF** (Open Shortest Path First).
- **BGP** (Border Gateway Protocol).

Table of figures

Figure 1-1 : BGP Routing [1].....1
Figure 1-3: BGP Peer Diagram [2].....2
Figure 1-5: BGP FSM [4].....3
Figure 2-1: Topology.....4
Figure 2-3.1: Configuring BGP on Router 1. 8
Figure 2-3.2: Configuring BGP on Router 2. 8
Figure 2-4.1: Define BGP Over OSPF on Router 1.....9
Figure 2-4.2: Define BGP Over OSPF on Router 2. 9
Figure 2-5.1: Define OSPF Over BGP on Router 1. 11
Figure 2-5.2: Define OSPF Over BGP on Router 2. 12
Figure 2-6: Configuring BGP Timers. 13
Figure 3-1.1: Show BGP Neighbors for Router 1. 14
Figure 3-1.2: Show BGP Neighbors for Router 2. 14
Figure 3-1.1: Show IP Route for Router 1. 15
Figure 3-1.2: Show IP Route for Router 2. 15
Figure 3-1.1: BGP neighbor summary for Router1.....18
Figure 3-1.2: BGP neighbor summary for Router 2.....18
Figure 3-1.1:BGP neighbor information for Router1.....18
Figure 3-2.1: BGP neighbor information for Router2.....18
Figure 3-2.1: Ping Between PC1 and PC3. 19
Figure 3-2.2: Ping Between PC1 and PC5. 20
Figure 3-2.3: Ping Between PC2 and PC3. 20
Figure 3-2.4: Ping Between PC2 and PC5. 21
Figure 3-2.5: Ping Between PC3 and PC1. 21
Figure 3-2.6: Ping Between PC3 and PC2. 22
Figure 3-2.8: Ping Between PC5 and PC1. 22
Figure 3-2.8: Ping Between PC5 and PC2. 23

List of tables

Table 2.1: Networks IPs. *5*

1. Introduction

1.1. Border Gateway Protocol (BGP)

The Border Gateway Protocol (BGP) is a crucial routing protocol used for inter-domain communication between Autonomous Systems (ASes), which are the building blocks of the Internet's routing infrastructure. Unlike Interior Gateway Protocols (IGPs) such as RIP, OSPF, and EIGRP, which function within a single AS, BGP is designed to route between multiple ASes. BGP Version 4 (BGPv4) is the current standard and utilizes a "Path Vector" approach, where routing decisions are based on AS paths and policies rather than metrics like distance or cost. This makes BGP essential for managing routing across large and diverse networks, supporting the Internet's global connectivity. [1][4].

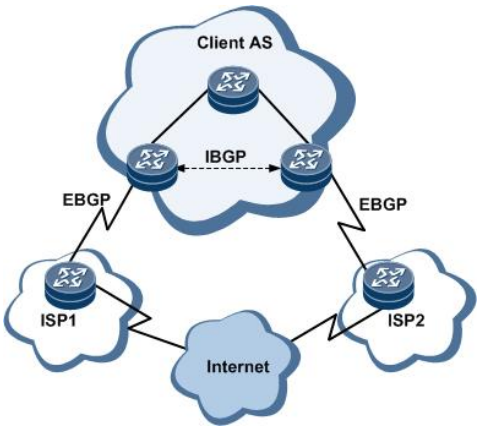


Figure 1-1: BGP Routing [1].

1.2. Using BGP

BGP is not always necessary for managing multiple Internet connections. For fault tolerance and redundancy, IGPs like OSPF or EIGRP are sufficient. BGP should only be used when multiple connections exist to external ASes, especially via different providers or separate routing policies. Additionally, BGP should be deployed when the existing infrastructure can handle its demands. Its key function is controlling incoming traffic to a local AS. [1].

To configure it, we use:

Router (config)# router bgp <AS-NUMBER>.

1.3. BGP Peers /Neighbors

For BGP to operate effectively, BGP routers (or "speakers") must establish neighbor relationships, also known as peers. These relationships can be either iBGP (within the same AS) or eBGP (between different ASes). After peers are connected, routers exchange their complete routing table, but only updates are transmitted moving forward. The administrative distance for eBGP routes is 20, while iBGP routes and locally originated routes have an administrative distance of 200. [2][3][4].

The configuration of an eBGP neighbor relationship can be done with:

Router(config-router)# neighbor <IP-ADDRESS-NEXT-INTERFACE> remote as <AS-OF-REMOTE-NEIGHBOR>.

Figure 1-3: BGP Peer Diagram [2].

1.4. BGP Peers Messages

In BGP, peer relationships are established and maintained through the exchange of specific message types, ensuring active connections and accurate routing information sharing:

- **OPEN**: Initiates the session, including parameters like BGP version, local AS number, and router ID. [5].
- **KEEPALIVE**: Sent periodically (every 60 seconds by default) to ensure the peer is reachable; if not received within the hold-time (180 seconds), the peer is considered unreachable. [5].
- **UPDATE**: Used to exchange routing information, advertising new routes or withdrawing outdated ones. [5].

- **NOTIFICATION**: Sent in case of fatal errors, resetting the BGP session when necessary. [5].

1.5. BGP Finite-State Machine (FSM)

BGP establishes peer sessions through a series of stages, ensuring connectivity and synchronization between routers. Once the session is confirmed, routing information is exchanged, enabling effective communication between networks. [5].

BGP FSM States:

- **Idle**: Initial state before connection attempts. [5].
- **Connect**: Waiting for a successful TCP connection. [5].
- **Active**: Attempting to establish the TCP connection. [5].
- **OpenSent**: TCP connection established; awaiting reply. [5].
- **OpenConfirm**: Listening for the peer's KEEPALIVE message. [5].
- **Established**: Session fully established; updates are exchanged. [5].

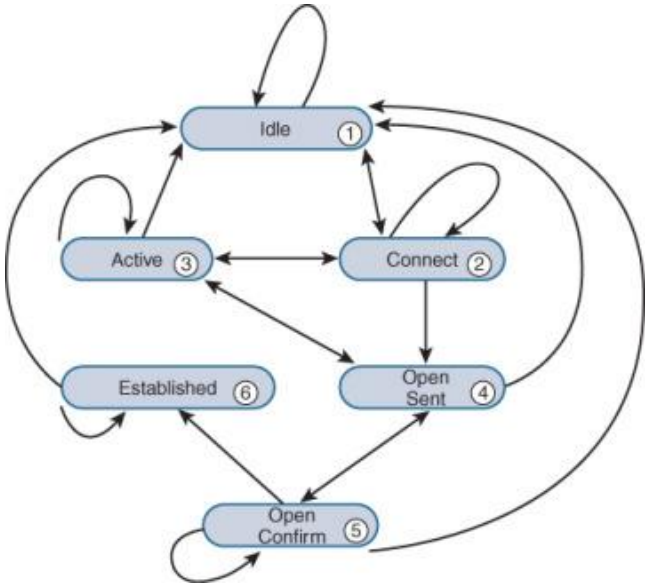


Figure 1-5: BGP FSM [4].

2. Procedure and Discussion

2.1. Building the Topology

The first step is to build the topology as shown in Figure 2-1 using Cisco Packet Tracer. This involves connecting four routers and several PCs across different networks. The routers will be represented using Router-PT, switches by Switch-PT, PCs by PC-PT and Sniffer to monitor packet exchanges. The connections between the devices should automatically use the appropriate connection type. The IP addresses for the devices are specified in Table 2-1, which will be used for configuration.

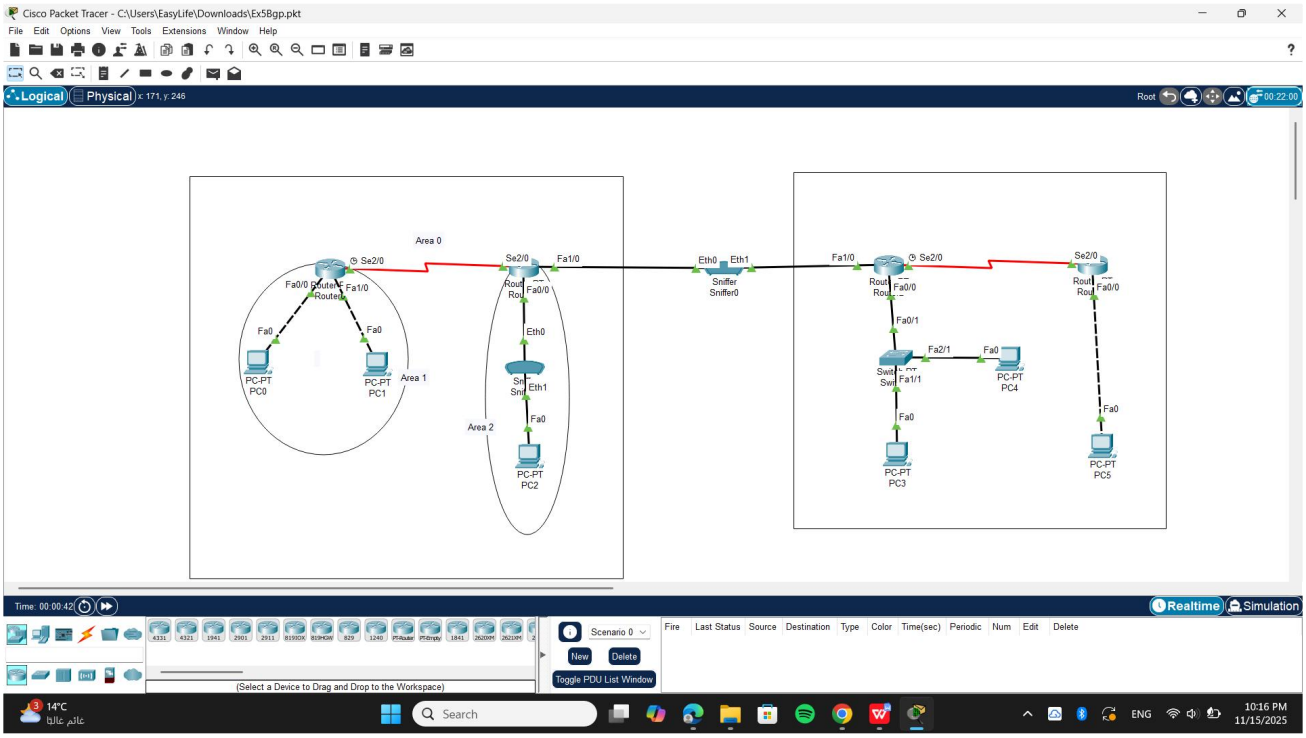


Figure 2-1: Topology.

Area/ AS & BGP Links	Network	Device	Interface	IP	Subnet Mask	Wildcard Mask
Area 0 / AS 100	192.74.0.0/30	Router 0	Se2/0	192. 74.0.1	255.255.255.252	0.0.0.3
		Router1	Se2/0	192. 74.0.2	255.255.255.252	0.0.0.3
Area 1 / As 100	192.74.1.32/27	Router 0	Fa1/0	192. 74.1.33	255.255.255.224	0.0.0.31
		PC1	Fa1	192. 74.1.34	255.255.255.224	0.0.0.31
	192.74.1.0/27	Router 0	Fa0/0	192. 74.1.1	255.255.255.224	0.0.0.31
		PC0	Fa0	192. 74.1.2	255.255.255.224	0.0.0.31
Area 2 / As 100	192. 74.1.64/26	Router 1	Fa0/0	192. 74.1.65	255.255.255.192	0.0.0.63
		PC2	Fa0	192. 74.1.66	255.255.255.192	0.0.0.63
AS 200	192. 74.1.192/26	Router 3	Fa0/0	192.74.1.193	255.255.255.192	0.0.0.63
		PC5	Fa0	192.74.1.194	255.255.255.192	0.0.0.63
	192. 74.1.128/26	Router 2	Fa0/0	192.74.1.129	255.255.255.192	0.0.0.63
		PC3	Fa0	192.74.1.130	255.255.255.192	0.0.0.63
		PC4	Fa0	192.74.1.131	255.255.255.192	0.0.0.63
BGP Links	192. 74.0.4/30	Router 1	Fa1/0	192. 74.0.5	255.255.255.252	0.0.0.3
		Router 2	Fa1/0	192. 74.0.6	255.255.255.252	0.0.0.3

Table 2.1-1Networks IPs.

2.2. Configuring OSPF Routing

In this step, we configure the OSPF routing protocol for Autonomous System 100 (AS 100), without including the BGP link in the OSPF configuration. The OSPF setup involves defining the router ID, enabling OSPF on the appropriate interfaces, and specifying the required network statements and area numbers for each router.

The following commands are used for configuring OSPF on each router:

- **Router 0 (AS 100):**

```
Router(config)# router ospf 1
Router(config-router)# network 192.74.0.0 0.0.0.3 area 0
Router(config-router)# network 192.74.1.0 0.0.0.31 area 1
Router(config-router)# network 192.74.1.32 0.0.0.31 area 1
```

- **Router 1 (AS 100):**

```
Router(config)# router ospf 1
Router(config-router)# network 192.74.0.0 0.0.0.3 area 0
Router(config-router)# network 192.74.1.64 0.0.0.63 area 2
```

2.3. Configuring EIGRP Routing

We will configure the EIGRP routing protocol for (AS 200), without including the Bgp link.

The following commands are used for configuring EIGRP on each router:

- **Router 3 (AS 200):**

```
Router ( config )# router eigrp 50
Router ( config - router )# network 192.74.1.192 0.0.0.63
Router ( config - router )# network 192.74.0.8 0.0.0.3
Router ( config - router )# no auto - summary
```

- **Router 2 (AS 200):**

Router (config)# router eigrp 50

Router (config - router)# network 192.74.1.128 0.0.0.63

Router (config - router)# network 192.74.0.8 0.0.0.3

Router (config - router)# no auto - summary

2.4. Configuring BGP Routing

BGP configuration needs to be performed only on Router 1 and Router 2, as they are responsible for connecting the different Autonomous Systems (AS 100 and AS 200). The first step in configuring BGP is to enable the BGP process and specify the router's Autonomous System (AS). This is done using the following command:

Router (config)# router bgp <AS-NUMBER>.

Once BGP is enabled on Router 1, the next step is to establish a neighbor relationship with a router in a separate AS (eBGP Peer). To do this, use the following command:

Router(config-router)# neighbor <IP-ADDRESS-NEXT-INTERFACE> remote as <AS-OF-REMOTE-NEIGHBOR>.

Where <IP-ADDRESS-NEXT-INTERFACE> is the IP address of the interface on the neighboring router, and <AS-OF-REMOTE-NEIGHBOR> is the Autonomous System number of the neighboring AS.

2.4.1. Configuring BGP Routing on Router 1

To configure BGP on Router 1, the following two commands are used:

1. **Enable BGP and specify the AS number (AS 100 for Router 1):**

Router(config)# router bgp 100

2. **Define the neighbor relationship with Router 2 (AS 200) using the remote AS and the IP address of Router 2's interface:**

Router(config-router)# neighbor 192.74.0.6 remote-as 200

2.4.2. Configuring BGP Routing on Router 2

To configure BGP on Router 2, the following two commands are used:

1. **Enable BGP and specify the AS number (AS 200 for Router 2):**

Router(config)# router bgp 200

2. **Define the neighbor relationship with Router 1 (AS 100) using the remote AS and the IP address of Router 1's interface:**

Router(config-router)# neighbor 192.74.0.5 remote-as 100

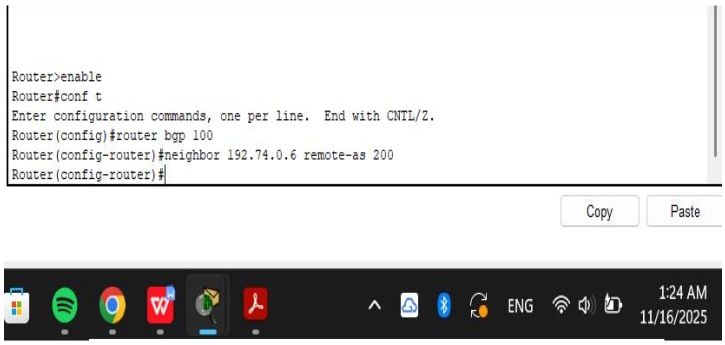


Figure 2-4.1: Configuring BGP on Router 1.

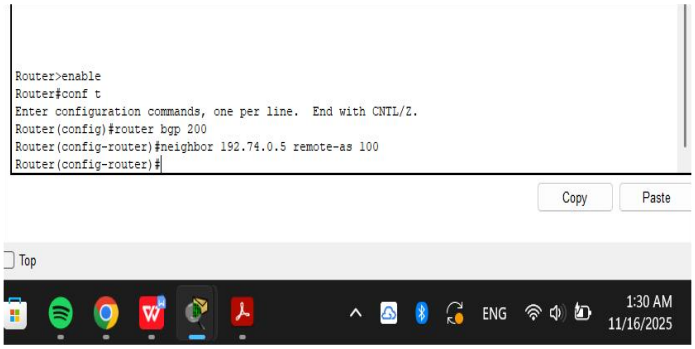


Figure 2-4.2: Configuring BGP on Router 2.

2.5. Redistributing BGP into OSPF

To enable communication between the OSPF and BGP protocols, a **redistribute** command is used. This command allows BGP routes to be included in the OSPF routing process.

The following commands are used to configure this:

1. **Enter the OSPF configuration mode:**

Router(config)# router ospf <PROCESS-ID>

Which, <PROCESS-ID> is the OSPF process ID.

2. **Redistribute BGP routes into OSPF:**

Router(config-router)# redistribute bgp <AS-NUMBER> subnets

In this command, <AS-NUMBER> refers to the Autonomous System number of the BGP configuration on the same router. The subnets keyword ensures that all BGP-learned subnets are redistributed into the OSPF process.

This configuration allows the OSPF protocol to communicate with and use the routing information learned from BGP, enabling inter-domain routing between different autonomous systems.

2.5.1. Define BGP Over OSPF in Router 1

To define BGP over OSPF in Router 1, use the following commands:

Router(config)# router ospf 1

Router(config-router)# redistribute bgp 100 subnets

2.5.2. Define BGP Over OSPF in Router 2

To define BGP over OSPF in Router 2, use the following commands:

Router(config)# router ospf 1

Router(config-router)# redistribute bgp 200 subnets

Figure 2-5.1: Define BGP Over OSPF on Router 1.

Figure 2-5.2: Define BGP Over OSPF on Router 2.

2.6. Redistributing OSPF into BGP

To allow communication from OSPF to BGP, a **redistribute** command is used to define OSPF over BGP. This allows BGP to learn routes from OSPF and vice versa. The commands used on the router are as follows:

1. **Enter the BGP configuration mode:**

Router(config)# router bgp <AS-NUMBER>

2. **Redistribute OSPF routes into BGP:**

Router(config-router)# redistribute ospf <PROCESS-ID>

2.6.1. Define OSPF Over BGP in Router 1

To define OSPF over BGP in Router 1, use the following commands:

Router(config)# router bgp 100

Router(config-router)# redistribute ospf 1

2.6.2. Define OSPF Over BGP in Router 2

To define OSPF over BGP in Router 2, use the following commands:

Router(config)# router bgp 200

Router(config-router)# redistribute ospf 1

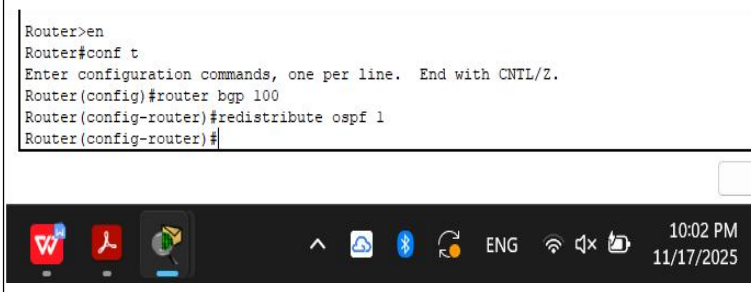


Figure 2-6.1: Define OSPF Over BGP on Router 1.

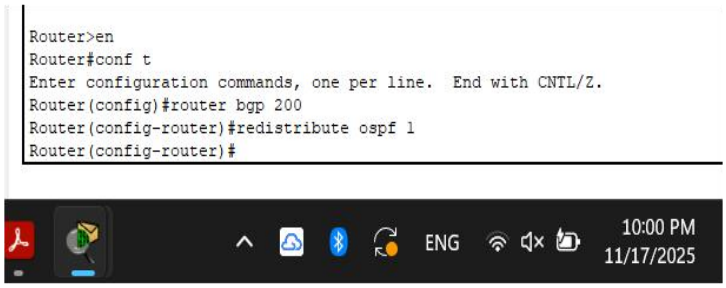


Figure 2-6.2: Define OSPF Over BGP on Router 2.

2.7. Redistributing BGP into EIGRP

When redistributing BGP routes into EIGRP, the metric values must match the interface through which the routes enter the EIGRP domain. In this topology, BGP routes enter through the Serial link between Router2 and Router3.

1. **Enter the EIGRP configuration mode:**

Router(config)# router eigrp <EIGRP-AS-NUMBER>

2. **Redistribute BGP routes into EIGRP:**

redistribute bgp <AS-NUMBER> metric <BW> <DELAY> <RELIABILITY> <LOAD> <MTU>

2.7.1. Define BGP into EIGRP in Router 2

To define BGP (AS 200) into EIGRP (AS 50) on Router2, use the following commands:

Router(config)# router eigrp 50

Router(config-router)# redistribute bgp 200 metric 1544 2000 255 1 1500

Figure 7.1-2: Define BGP into EIGRP on Router 2

2.8. Redistributing EIGRP into BGP

To allow BGP to learn routes from the EIGRP domain, a redistribution command is required. Since Router2 runs both BGP AS 200 and EIGRP AS 50, it is the router where redistribution must be configured.

1. **Enter the BGP configuration mode:**

Router(config)# router bgp <AS-NUMBER>

2. **Redistribute EIGRP routes into BGP:**

Router(config-router)# redistribute eigrp <EIGRP-AS-NUMBER>

2.8.1. Define EIGRP into BGP in Router 2

To define EIGRP (AS 50) into BGP (AS 200) on Router2, use the following commands:

Router(config)# router bgp 200

Router(config-router)# redistribute eigrp 50

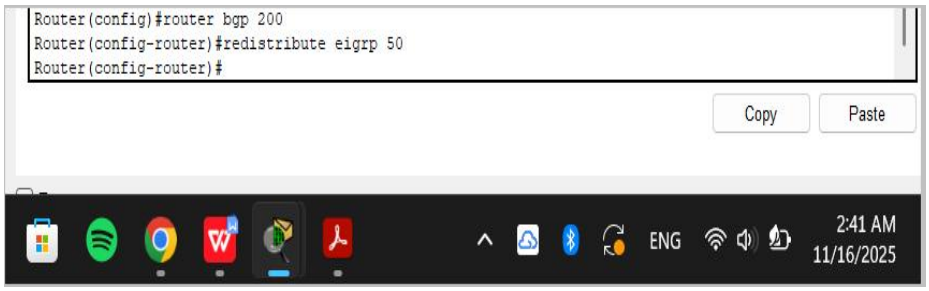


Figure 2-8.1: Define EIGRP into BGP on Router 2.

2.9. Configuring BGP Timers

BGP timers control the frequency of keepalive messages and the maximum time before a peer is considered down. To set the **keepalive** timer to 30 seconds and the **hold time** to 90 seconds on Router 1, use the command timers bgp 30 90. If peers have different timer settings, the session will still establish, using the smaller timer value.

To configure the BGP keepalive and hold time for Router 1, we used the following commands:

Router(config)# router bgp 100

Router(config-router)# timers bgp 30 90

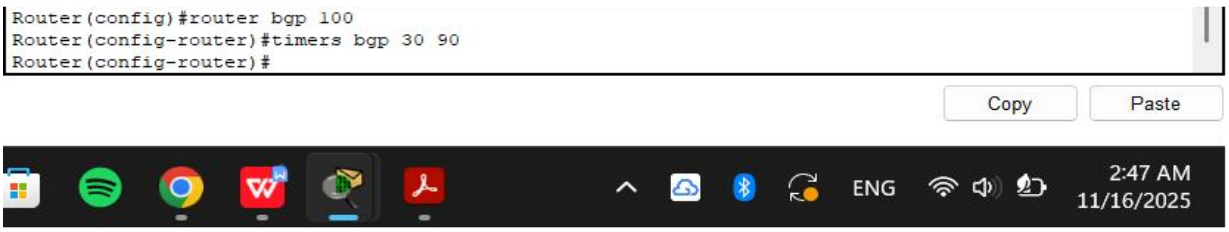


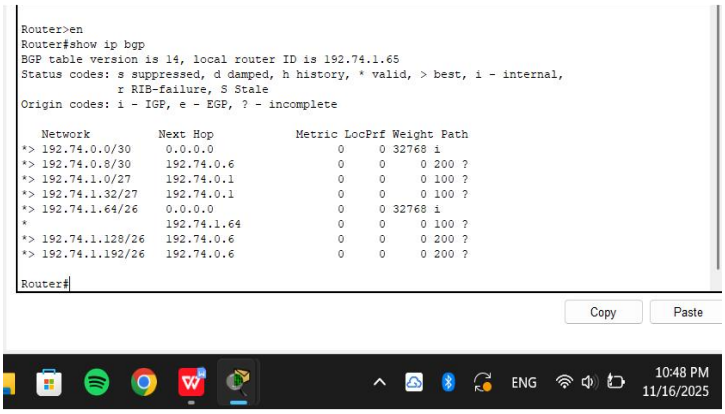
Figure 2-9: Configuring BGP Timers.

3. Results

3.1. Viewing BGP Neighbors

To verify and monitor the status of all BGP neighbors, the following commands were utilized:

- **show ip bgp (Viewing the BGP Routing Table)**: Displays detailed information about all BGP routes, including their path attributes and next-hop IP addresses.

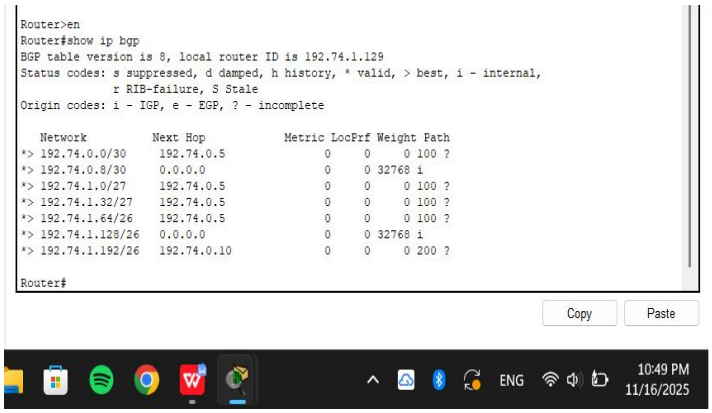


```
Router>en
Router#show ip bgp
BGP table version is 14, local router ID is 192.74.1.65
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 192.74.0.0/30     0.0.0.0                  0     0 32768 i
*> 192.74.0.8/30     192.74.0.6               0     0     0 200 ?
*> 192.74.1.0/27     192.74.0.1               0     0     0 100 ?
*> 192.74.1.32/27    192.74.0.1               0     0     0 100 ?
*> 192.74.1.64/26    0.0.0.0                  0     0 32768 i
*                    192.74.1.64              0     0     0 100 ?
*> 192.74.1.128/26   192.74.0.6               0     0     0 200 ?
*> 192.74.1.192/26   192.74.0.6               0     0     0 200 ?

Router#
```

Figure 3-1.1: Show BGP Neighbors for Router 1.



```
Router>en
Router#show ip bgp
BGP table version is 8, local router ID is 192.74.1.129
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 192.74.0.0/30     192.74.0.5               0     0     0 100 ?
*> 192.74.0.8/30     0.0.0.0                  0     0 32768 i
*> 192.74.1.0/27     192.74.0.5               0     0     0 100 ?
*> 192.74.1.32/27    192.74.0.5               0     0     0 100 ?
*> 192.74.1.64/26    192.74.0.5               0     0     0 100 ?
*> 192.74.1.128/26   0.0.0.0                  0     0 32768 i
*> 192.74.1.192/26   192.74.0.10              0     0     0 200 ?

Router#
```

Figure 3-1.2: Show BGP Neighbors for Router 2.

- **BGP Table Overview for Router 1:**

- **BGP Table Version**: 14 – Indicates the number of changes to the table since BGP started on this router.
- **Local Router ID**: 192.74.1.65 – The identifier for Router 1 in the BGP system.

- **Status Codes**

- *> indicates the best route (> indicates it is preferred) and is valid (*).

- **Route Analysis**

1. **192.74.0.0/30**

   - Best route (denoted by *>) with a next-hop of 0.0.0.0, indicating it is directly connected.
   - Alternate entry with ? shows the route learned from AS 100 but not preferred.

2. **192.74.0.8/30**

- Best route via 192.74.0.6, indicating the next-hop address is through Router 2.

3. **192.74.1.0/27 and 192.74.1.32/27**

- Best routes via 192.74.0.1, which is likely a connected interface R0 to another router in AS 100.

4. **192.74.1.64/26**

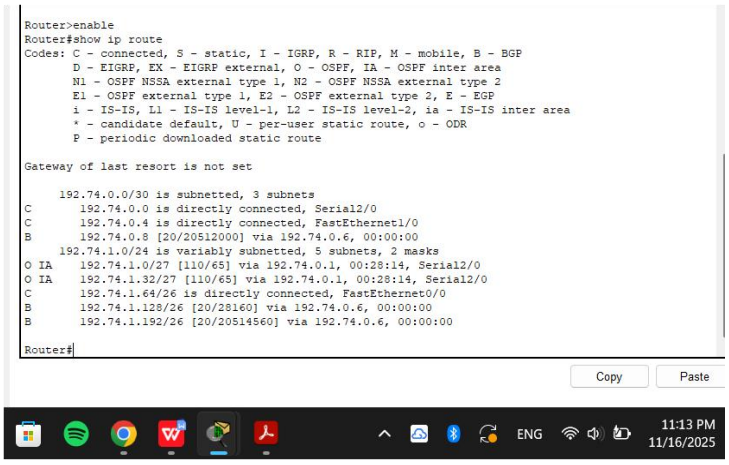
- Best route is directly connected (0.0.0.0, origin: i).

5. **192.74.1.128/26 and 192.74.1.192/26**

- Best routes via 192.74.0.6, originating from AS 200 (eBGP peers).

This table validates that Router 1 is effectively participating in BGP, maintaining accurate neighbor relationships and selecting optimal routes for traffic forwarding.

- **show ip route**: Shows the overall routing table, indicating how BGP routes integrate with other protocols.



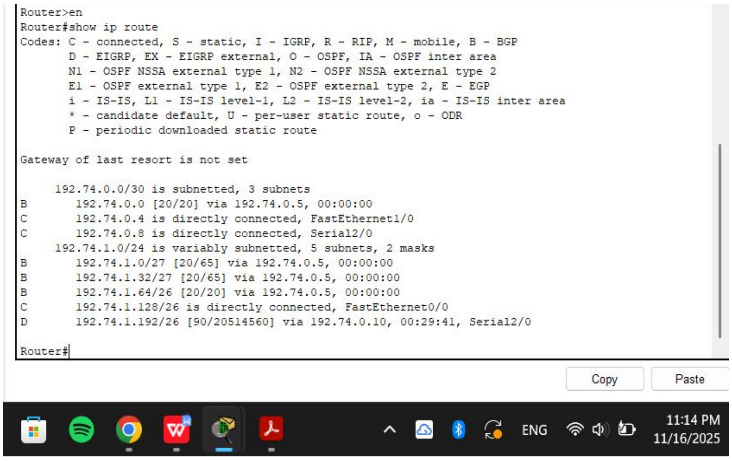
```
Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.74.0.0/30 is subnetted, 3 subnets
C       192.74.0.0 is directly connected, Serial2/0
C       192.74.0.4 is directly connected, FastEthernet1/0
B       192.74.0.8 [20/20512000] via 192.74.0.6, 00:00:00
     192.74.1.0/24 is variably subnetted, 5 subnets, 2 masks
O IA    192.74.1.0/27 [110/65] via 192.74.0.1, 00:28:14, Serial2/0
O IA    192.74.1.32/27 [110/65] via 192.74.0.1, 00:28:14, Serial2/0
C       192.74.1.64/26 is directly connected, FastEthernet0/0
B       192.74.1.128/26 [20/28160] via 192.74.0.6, 00:00:00
B       192.74.1.192/26 [20/20514560] via 192.74.0.6, 00:00:00

Router#
```

Figure 3-1.1: Show IP Route for Router 1.



```
Router>en
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.74.0.0/30 is subnetted, 3 subnets
B       192.74.0.0 [20/20] via 192.74.0.5, 00:00:00
C       192.74.0.4 is directly connected, FastEthernet1/0
C       192.74.0.8 is directly connected, Serial2/0
     192.74.1.0/24 is variably subnetted, 5 subnets, 2 masks
B       192.74.1.0/27 [20/65] via 192.74.0.5, 00:00:00
B       192.74.1.32/27 [20/65] via 192.74.0.5, 00:00:00
B       192.74.1.64/26 [20/20] via 192.74.0.5, 00:00:00
C       192.74.1.128/26 is directly connected, FastEthernet0/0
D       192.74.1.192/26 [90/20514560] via 192.74.0.10, 00:29:41, Serial2/0

Router#
```

Figure 3-1.2: Show IP Route for Router 2.

- **Analysis of the show ip route Command on Router 1**

1. Route Types and Protocols

- The table includes routes learned through:
  - **C** (Connected): Directly connected networks.
  - **O** (OSPF): Routes learned through the OSPF dynamic routing protocol.
  - **B** (BGP): Routes learned through Border Gateway Protocol.

---

2. Subnetted Networks

192.74.0.0/30

- **Subnets (3):**
  - 192.74.0.0 and 192.74.0.4: Directly connected via Serial2/0 and FastEth1/0, respectively.
  - **BGP Learned**: 192.74.0.8 with a metric [20/20512000] via 192.74.0.6. The first 20 is the administrative distance (AD) for BGP, and the second 20512000 is the metric.

192.70.1.0/24

- **Subnets (5):**
  - **OSPF Routes**:
    - 192.74.1.0/27 and 192.74.1.32/27: Learned via OSPF (O) with AD [110] and cost 65, reachable through 192.74.0.1 over Serial2/0.
  - **Directly Connected**: 192.74.1.64/26 through FastEthernet0/0.
  - **BGP Learned**:
    - 192.74.1.128/26 and 192.74.1.192/26: Learned via BGP (B) with AD [20]. Both routes are reachable through 192.74.0.6.

---

3. Key Observations

1. **Connected Routes**:

- All directly connected routes (C) show proper interface mapping (e.g., Serial2/0, FastEthernet1/0, FastEthernet0/0).
- These are immediately available to Router 1 without needing dynamic protocols.

2. **OSPF Routes**:

- The OSPF-learned routes (O) for 192.74.1.0/27 and 192.74.1.32/27 are correctly propagated with AD [110].
- Both routes use 192.74.0.1 as the next hop.

3. **BGP Routes**:

- BGP-learned routes (B) for 192.74.0.8, 192.74.1.128/26, and 192.74.1.192/26 are correctly populated.
- The AD of [20] indicates these are external BGP (eBGP) routes.
- The consistent next-hop address (192.74.0.6) confirms proper eBGP neighbor configuration.

4. **Administrative Distance (AD)**:

- Connected routes: AD 0 (highest priority).
- OSPF routes: AD 110 (intermediate priority).
- BGP routes: AD 20 (lower priority for eBGP, but higher than OSPF).

5. **Interface Utilization**:

- OSPF routes are learned over Serial interfaces, while FastEthernet is used for directly connected routes (192.74.1.64/26).
- BGP routes use FastEthernet for the next hop (192.74.0.6).

- **show ip bgp summary**: Provides a concise summary of BGP neighbors, their current state, and session details.

```
Router#show ip bgp summary
BGP router identifier 192.74.1.65, local AS number 100
BGP table version is 23, main routing table version 6
9 network entries using 1188 bytes of memory
9 path entries using 468 bytes of memory
3/3 BGP path/bestpath attribute entries using 552 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2288 total bytes of memory
BGP activity 7/0 prefixes, 9/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.74.0.6      4   200      32      29       23    0    0 00:13:49        4

Router#
```

Figure 3-1.1: BGP neighbor summary for Router1.

```
Router#show ip bgp summary
BGP router identifier 192.74.1.129, local AS number 200
BGP table version is 11, main routing table version 6
7 network entries using 924 bytes of memory
7 path entries using 364 bytes of memory
4/4 BGP path/bestpath attribute entries using 736 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2104 total bytes of memory
BGP activity 7/0 prefixes, 7/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.74.0.5      4   100      47      33       11    0    0 00:15:45        4

Router#
```

Figure 3-1.2: BGP neighbor summary for Router 2.

- **show ip bgp neighbors:** provides detailed information about BGP neighbors, including the remote AS number, BGP version, session state, and timers, and is used to verify neighbor relationships and ensure stable connections.

```
Router#show ip bgp neighbors
BGP neighbor is 192.74.0.6,  remote AS 200, external link
  BGP version 4, remote router ID 192.74.1.129
  BGP state = Established, up for 00:22:35
  Last read 00:22:35, last write 00:22:35, hold time is 90, keepalive interval is 30 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:               10          3
    Keepalives:            46         46
    Route Refresh:          0          0
    Total:                 57         50
  Default minimum time between advertisements runs is 30 seconds

 For address family: IPv4 Unicast
  BGP table version 23, neighbor version 6/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
    Prefixes Current:               6          3 (Consumes 207 bytes)
    Prefixes total:                 6          3
    Implicit Withdraw:              0          0
    Explicit Withdraw:              0          0
    Used as bestpath:             n/a          1
    Used as multipath:            n/a          0

                                   Outbound    Inbound
     Local Policy Denied Prefixes:    --------    -------
       Total:                             0           0
     Number of NLRIs in the update sent: max 3, min 1
```

Figure 3-1.1:BGP neighbor information for Router1.

```
Router#show ip bgp neighbors
BGP neighbor is 192.74.0.5,  remote AS 100, external link
  BGP version 4, remote router ID 192.74.1.65
  BGP state = Established, up for 00:23:46
  Last read 00:23:46, last write 00:23:46, hold time is 90, keepalive interval is 30 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                3         14
    Keepalives:            48         48
    Route Refresh:          0          3
    Total:                 52         66
  Default minimum time between advertisements runs is 30 seconds

 For address family: IPv4 Unicast
  BGP table version 11, neighbor version 6/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
    Prefixes Current:               3          4 (Consumes 161 bytes)
    Prefixes total:                 3          4
    Implicit Withdraw:              0          0
    Explicit Withdraw:              0          0
    Used as bestpath:             n/a          1
    Used as multipath:            n/a          0

                                   Outbound    Inbound
     Local Policy Denied Prefixes:    --------    -------
       Total:                             0           0
     Number of NLRIs in the update sent: max 3, min 1

  Address tracking is enabled, the RIB does have a route to 192.74.0.5
```

Figure 3-2.1: BGP neighbor information for Router2.

3.2. Ping Between Devices

Ping is a vital network diagnostic tool used to test the connectivity between two devices. It works by sending **ICMP Echo Request** packets to the target device and waiting for **Echo Reply** responses. The **round-trip time (RTT)** is calculated to measure the time taken for the packets to travel to the destination and back.

This tool aids network administrators by:

- Verifying connectivity between devices.
- Evaluating network conditions, such as latency.
- Detecting issues like **packet loss** or high latency.
- Ensuring devices are properly configured and operational.

3.2.1. Ping Between PC1 (192.74.1.34) and PC3 (192.74.1.130)

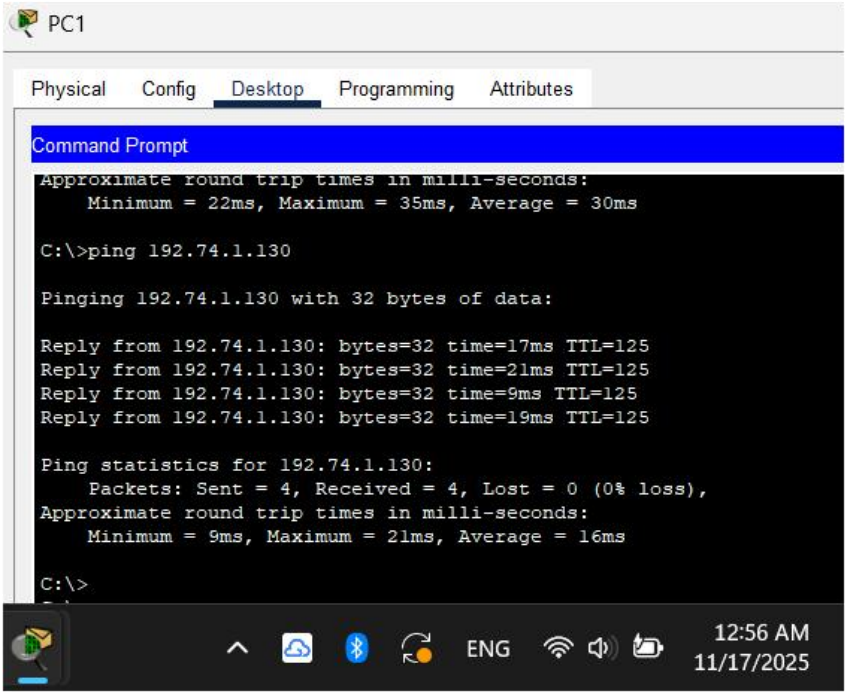


Figure 3-2.1: Ping Between PC1 and PC3.

The ping test confirms successful BGP connectivity between devices, with 0% packet loss and minimal latency.

3.2.2. Ping Between PC1 (192.74.1.34) and PC5 (192.74.1.194)

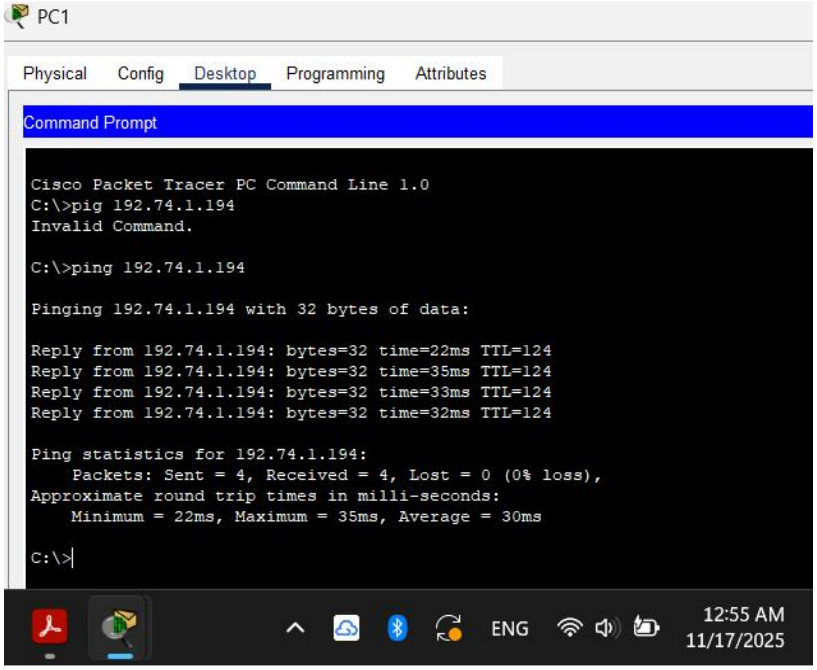


Figure 3-2.2: Ping Between PC1 and PC5.

3.2.3. Ping Between PC2 (192.74.1.66) and PC3 (192.74.1.130)

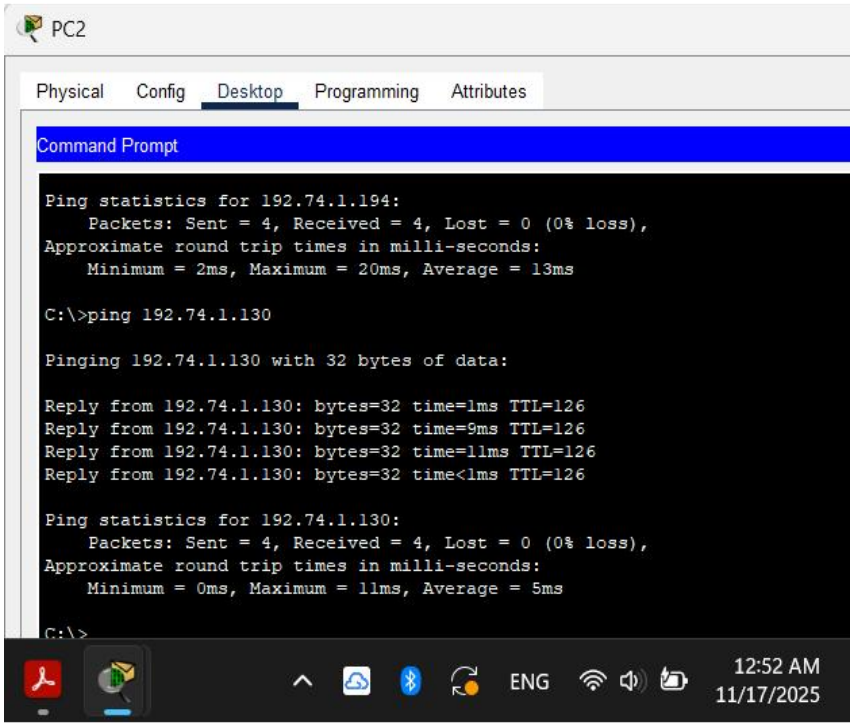


Figure 3-2.3: Ping Between PC2 and PC3.

3.2.4. Ping Between PC2 (192.74.1.66) and PC5 (192.74.1.194)

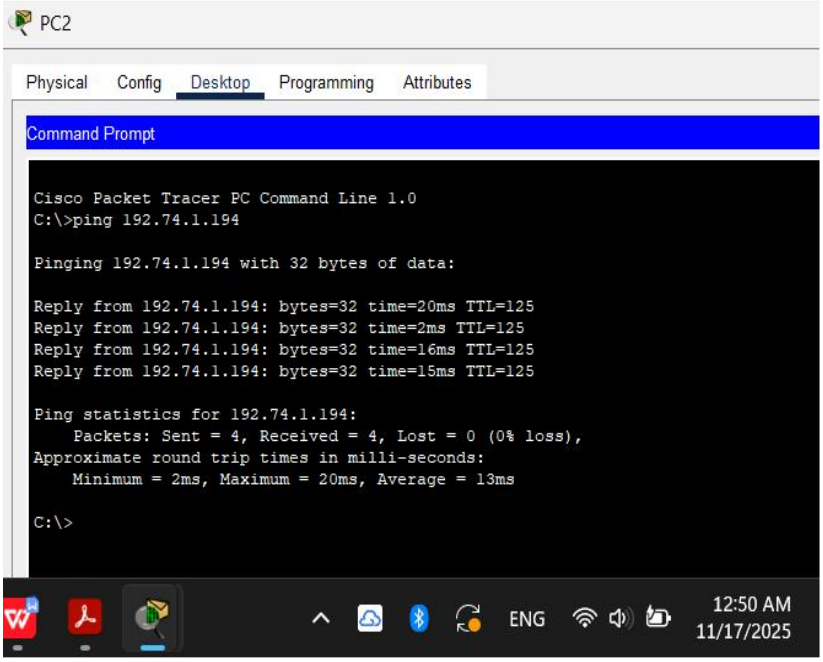


Figure 3-2.4: Ping Between PC2 and PC5.

3.2.5. Ping Between PC3 (192.74.1.130) and PC1 (192.74.1.34)

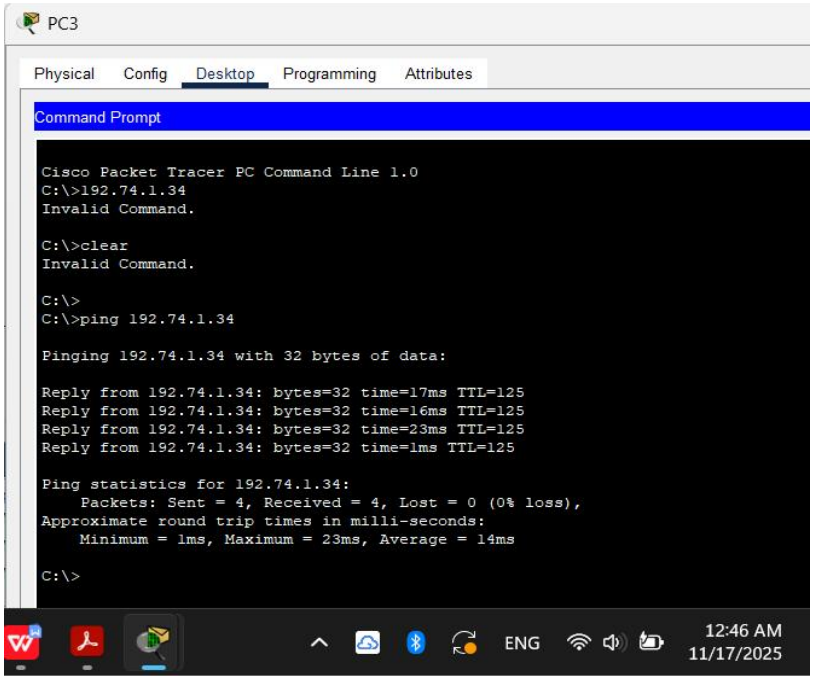


Figure 3-2.5: Ping Between PC3 and PC1.

3.2.6. Ping Between PC3 (192.74.1.130) and PC2 (192.74.1.66)

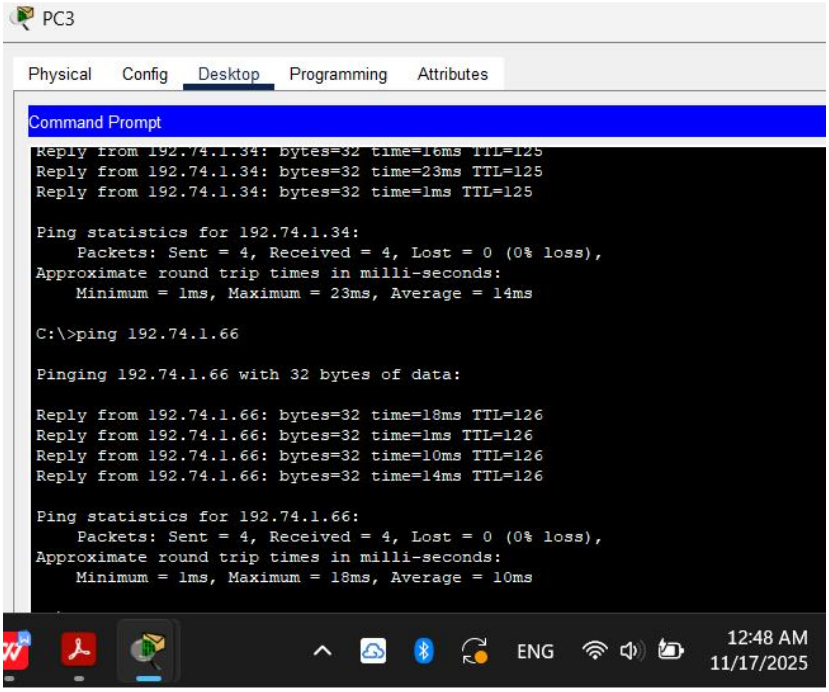


Figure 3-2.6: Ping Between PC3 and PC2.

3.2.7. Ping Between PC5 (192.74.1.194) and PC1 (192.74.1.34)

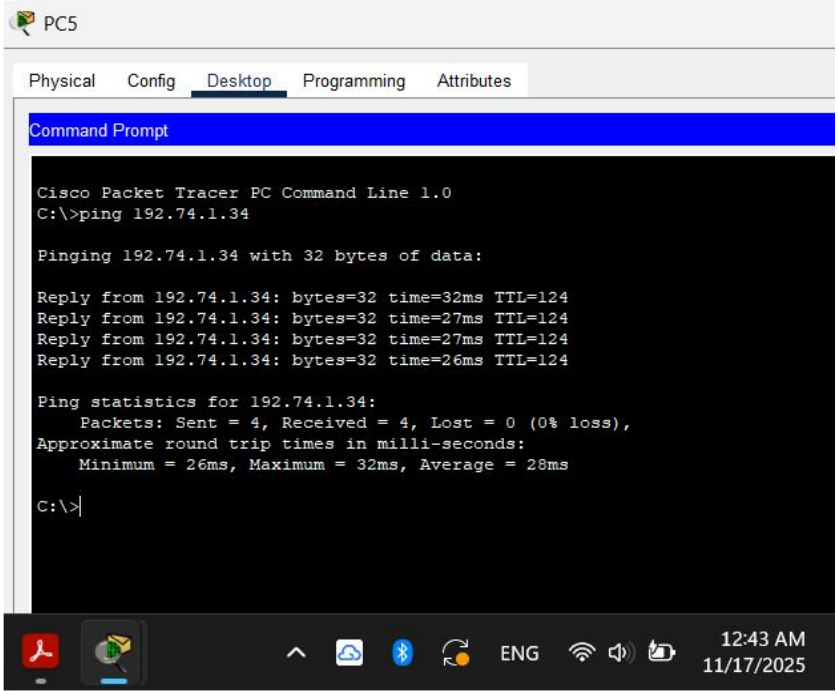


Figure 3-2.7: Ping Between PC5 and PC1.

The ping test confirms successful BGP connectivity between devices, with 0% packet loss and minimal latency.

3.2.8. Ping Between PC5 (192.74.1.194) and PC2 (192.74.1.66)

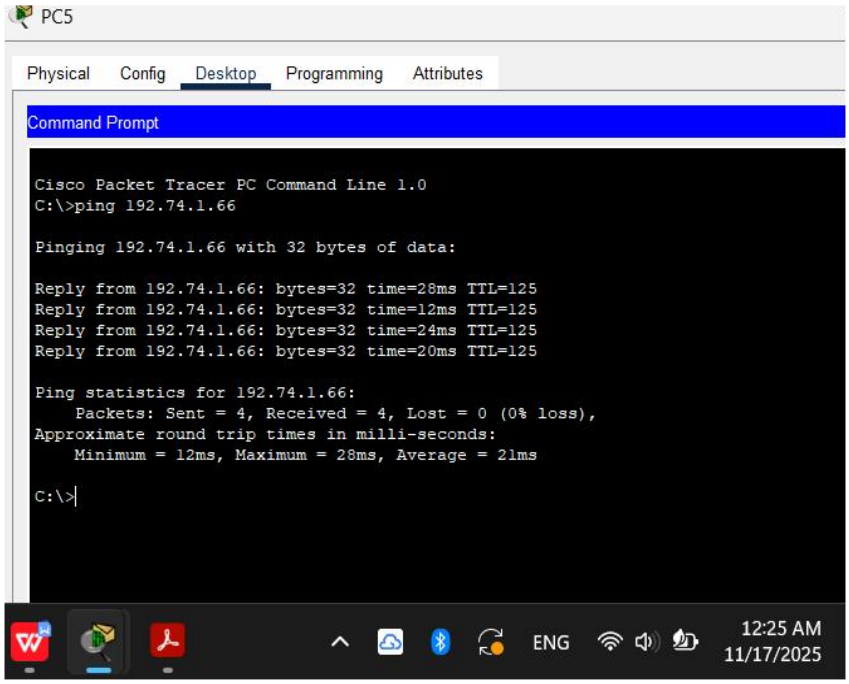


Figure 3-2.8: Ping Between PC5 and PC2.

The results confirm the successful implementation and testing of BGP. All routing tables were correctly updated, BGP neighbors were established without errors, and the ping tests demonstrated effective connectivity with no packet loss and minimal latency. This highlights the proper configuration of BGP protocols, seamless communication between devices, and reliable routing functionality throughout the network.

Conclusion

In this experiment, we configured OSPF, EIGRP, and BGP using Cisco Packet Tracer. The main objectives—setting up OSPF and EIGRP within their ASes, establishing BGP between ASes, and redistributing routes—were successfully achieved. Connectivity and routing were verified using Ping and routing tables. The lab confirmed that OSPF and EIGRP handle internal routing, BGP manages external routing, and redistribution enables communication across different protocols. Overall, the experiment reinforced the importance of proper protocol configuration and route redistribution for efficient network operation.

Feedback

I am pleased with the outcome of the experiment as it successfully achieved its objectives of configuring BGP and understanding its role in routing between autonomous systems. The time allocated for the experiment was sufficient to cover all the key sections, allowing for a thorough exploration of both OSPF, EIGRP and BGP. The hands-on experience was both educational and rewarding, offering deep insights into BGP's scalability and its crucial role in inter-domain routing. Additionally, the experiment helped clarify the differences between interior and exterior gateway protocols and provided a practical understanding of policy-based routing. Overall, it was a valuable learning experience that enhanced my understanding of dynamic routing in large-scale networks.

References

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[2]: Geeksforgeeks. Border Gateway Protocol (BGP) - GeeksforGeeks [Accessed on 13-Nov-2025]

[3]: catchpoint. The Guide to Dynamic Routing Protocols [Accessed on 14-Nov-2025]

[4] : Cisco Press. BGP Neighbor States > BGP Fundamentals | Cisco Press. (n.d.). https://www.ciscopress.com/articles/article.asp?p=2756480&seqNum=4 [Accessed on 15-Nov-2025]

[5]: BGP - Finite State Machine (FSM). NetworkLessons Notes. (n.d.). https://notes.networklessons.com/bgp-finite-state-machine-fsm [Accessed on 13-Nov-2025]